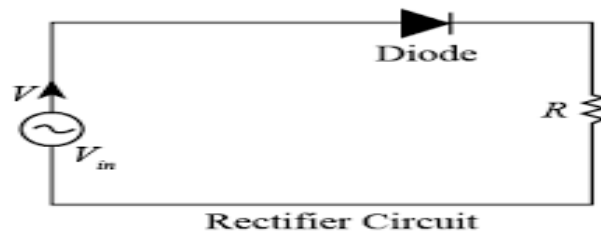


Expt. No.1

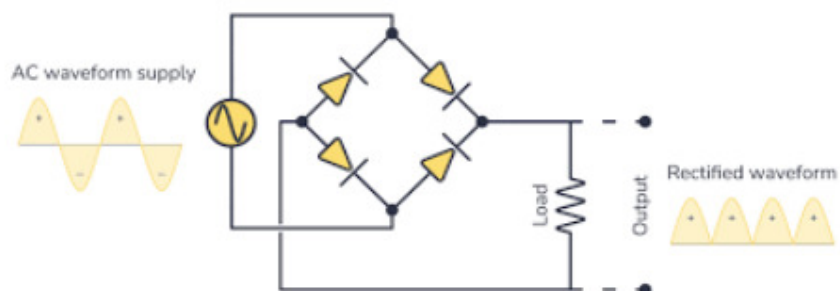
Simulation of diode rectifier

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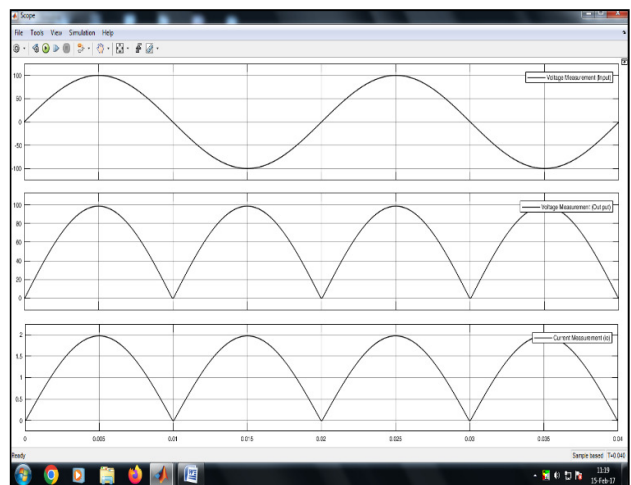
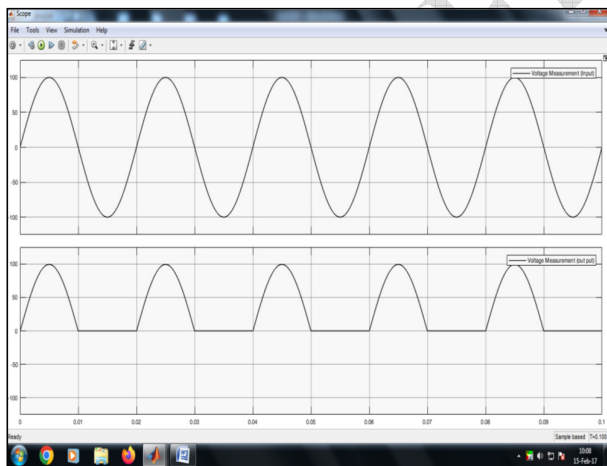
CIRCUIT DIAGRAM of HWR



CIRCUIT DIAGRAM of FWR



NATURE OF GRAPH:



Expt.No.1

Simulation of Diode rectifier

AIM:

- 1) Simulate a single phase half wave diode bridge rectifier. Input 100V, 50 Hz. AC supply. At the output, resistance of 50 ohms.
- 2) Simulate a single phase full wave diode bridge rectifier. Input 100V, 50 Hz. AC supply. At the output, resistance of 50 ohms.

THEORY:

PROCEDURE:

- 1) Open Matlab tool using icon on the desktop and click on Simulink icon in the home tab.
- 2) Click on new blank module and select the required components from the library function and drop on the work space.
- 3) Connect all components as per the circuit diagram
- 4) Use scope to observe input and output voltages and currents.
- 5) Save the file with a suitable name and set the run time in the menu bar to 0.1sec
- 6) Run the simulation and observe the waveforms in the scope.
- 7) Measure the value of output quantities using measurement options in the scope.
- 8) Note down the values in tabular column.

Tabular Column:

VM(Max)	Vdc	Idc	Vrms	Efficiency $\eta = V_{dc}^2/V_{rms}^2$

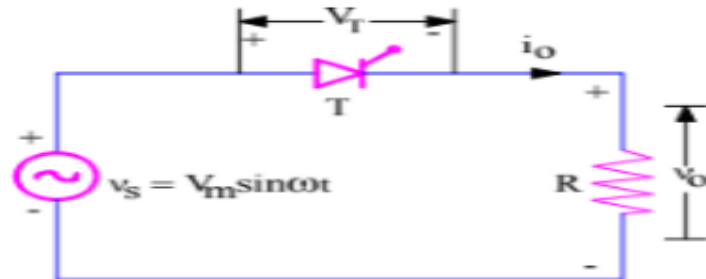
Result:

Conclusion:

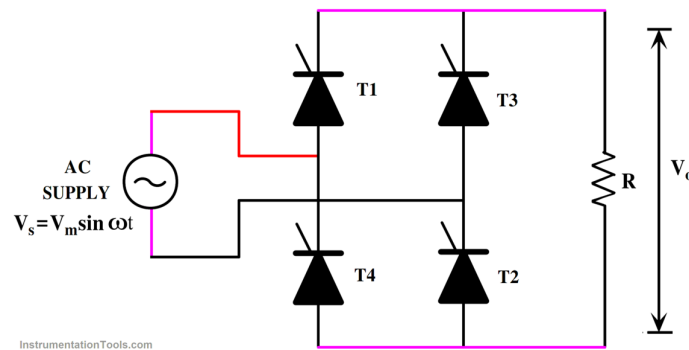
Expt. No.2

Simulation of Controlled rectifier

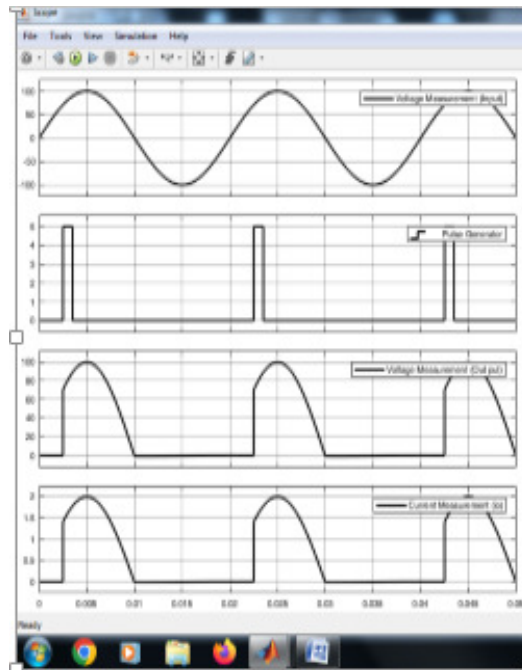
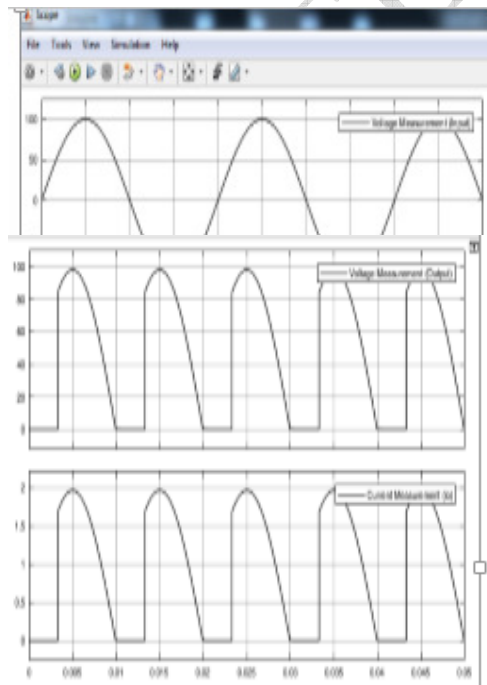
CIRCUIT DIAGRAM of HWR



CIRCUIT DIAGRAM of FWR



NATURE OF GRAPH:



Expt.No.2

Simulation of Controlled rectifier

AIM:

Simulate a buck converter with 20 V DC input, and regulate the output at 10 V by implementing a PI controller for closed loop operation. The output power to vary from 10 W to 20 W. Ensure that voltage ripple is limited to 1%.

THEORY:

PROCEDURE:

- 1) Open Matlab tool using icon on the desktop and click on Simulink icon in the home tab.
- 2) Click on new blank module and select the required components from the library function and drop on the work space.
- 3) Connect all components as per the circuit diagram
- 4) Use scope to observe input and output voltages and currents.
- 5) Save the file with a suitable name and set the run time in the menu bar to 0.1sec
- 6) Run the simulation and observe the waveforms in the scope.
- 7) Measure the value of output quantities using measurement options in the scope.
- 8) Note down the values in tabular column.

Tabular Column:

Firing angle α	Pulse Gate-1 Delay Time in sec	Pulse Gate-2 Delay Time in sec	Output Voltage Mean	Output Voltage RMS	Output I_o Mean	Output I_o RMS

Result:

Conclusion:

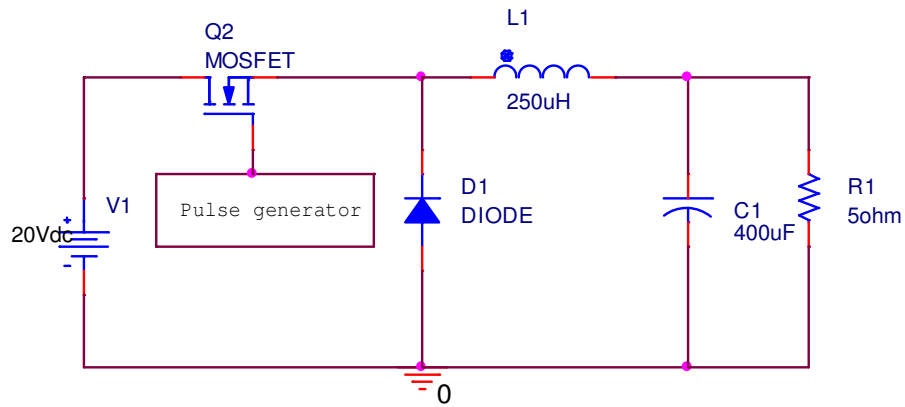
Expt. No.3

Simulation of Buck converter

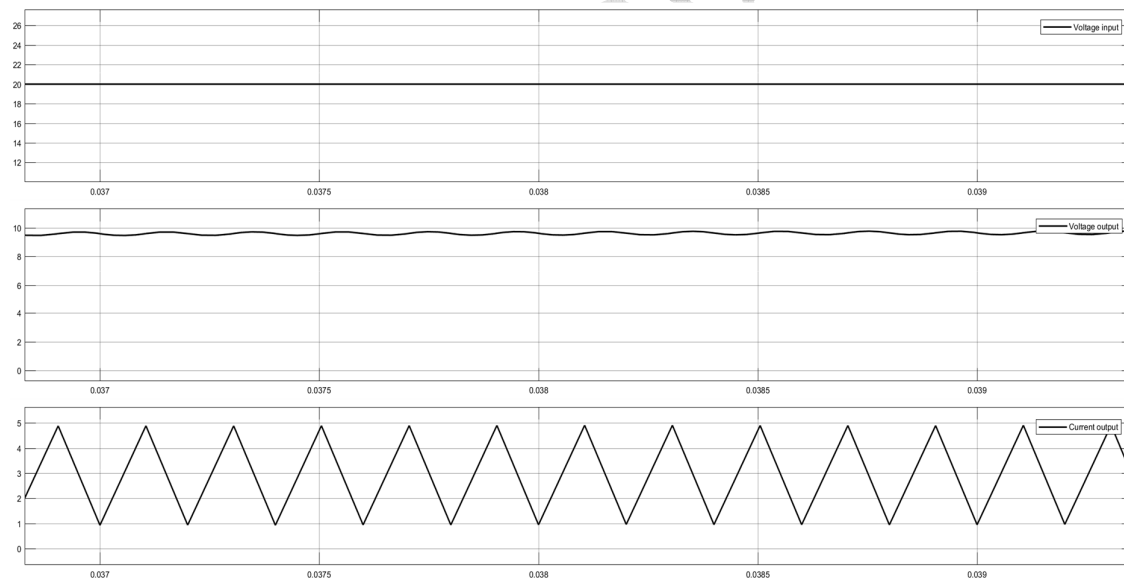
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Circuit Diagram:

Buck converter



Nature of waveform:



Expt.No.3

Simulation of Buck Converter

AIM:

Simulate a buck converter with 20 V DC input, and regulate the output at 10 V by implementing a PI controller for closed loop operation. The output power to vary from 10 W to 20 W. Ensure that voltage ripple is limited to 1%.

THEORY:

PROCEDURE:

- 1) Open Matlab tool using icon on the desktop and click on Simulink icon in the home tab.
- 2) Click on new blank module and select the required components from the library function and drop on the work space.
- 3) Connect all components as per the circuit diagram
- 4) Set $P=0.01$ & $I=5$ in PI controller module. Set $f=5000\text{Hz}$ in PWM generator module.
- 5) Use scope to observe input and output voltages and currents.
- 6) Save the file with a suitable name and set the run time in the menu bar to 0.1sec
- 7) Run the simulation and observe the waveforms in the scope.
- 8) Measure the value of output quantities using measurement options in the scope.
- 9) Note down the values in tabular column.

Tabular Column:

Sl. No.	Load current (I_o)	Output voltage (V_o)	Output power $P=V_o*I_o$

Result:

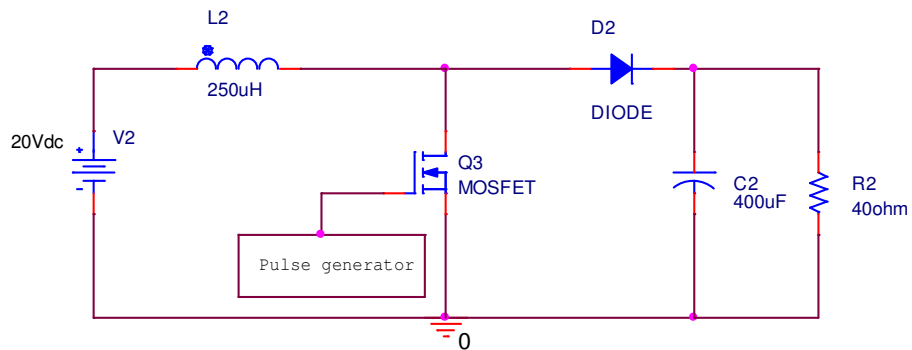
Conclusion:

Expt. No.4

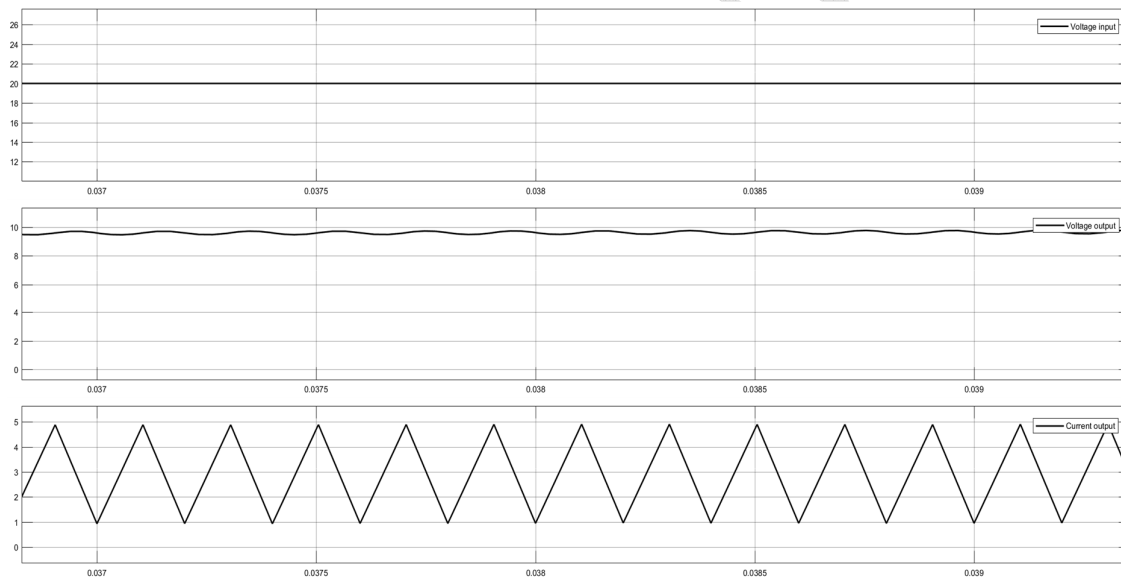
Simulation of Boost converter

Circuit Diagram:

Boost converter



Nature of waveform:



Expt.No.4

Simulation of Boost Converter

AIM:

Simulate a boost converter with 20 V DC input, and regulate the output at 35 V by implementing a PI controller for closed loop operation. The output power to vary from 30W to 60 W. Ensure that voltage ripple is limited to 1%

THEORY:

PROCEDURE:

- 1) Open Matlab tool using icon on the desktop and click on Simulink icon in the home tab.
- 2) Click on new blank module and select the required components from the library function and drop on the work space.
- 3) Connect all components as per the circuit diagram
- 4) Set $P=0.01$ & $I=5$ in PI controller module. Set $f=25000\text{Hz}$ in PWM generator module.
- 5) Use scope to observe input and output voltages and currents.
- 6) Save the file with a suitable name and set the run time in the menu bar to 0.1sec
- 7) Run the simulation and observe the waveforms in the scope.
- 8) Measure the value of output quantities using measurement options in the scope.
- 9) Note down the values in tabular column.

Tabular Column:

Sl. No.	Load current (I_o)	Output voltage (V_o)	Output power $P=V_o \cdot I_o$

Result:

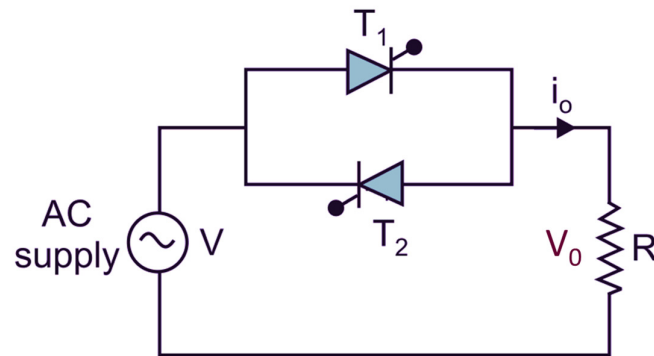
Conclusion:

Expt. No.5

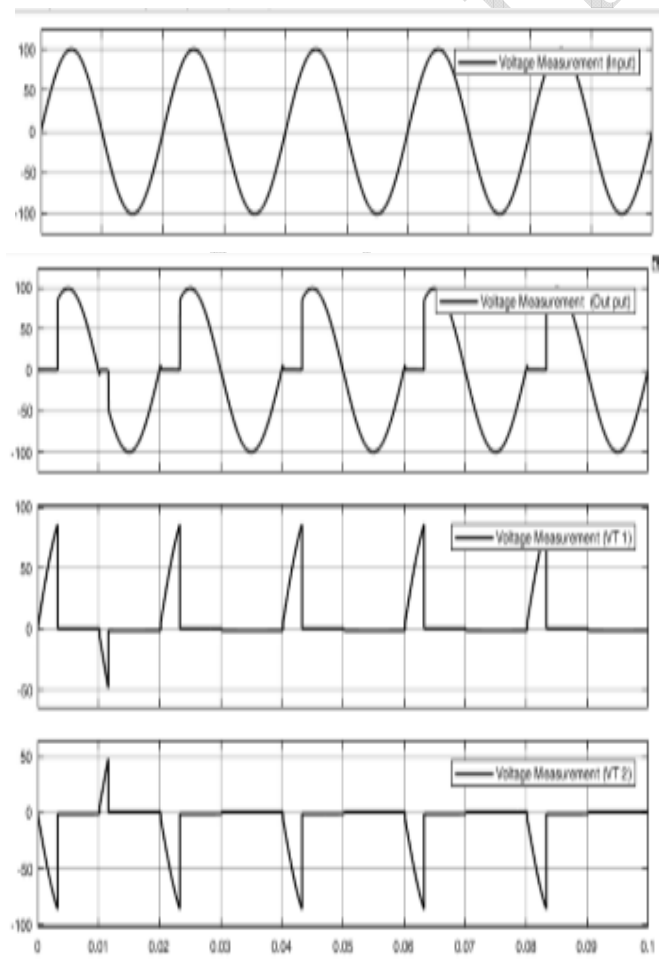
Simulation of Single phase AC voltage controller

Circuit Diagram:

AC voltage controller



Nature of waveform:



Expt.No.5

Simulation of AC voltage controller

AIM:

Simulate a single phase AC voltage controller using a triac with 100V, 50 Hz. AC supply for an RL load of 10 ohms and 2 mH.

THEORY:

PROCEDURE:

- 1) Open Matlab tool using icon on the desktop and click on Simulink icon in the home tab.
- 2) Click on new blank module and select the required components from the library function and drop on the work space.
- 3) Connect all components as per the circuit diagram
- 4) Vary firing angle in steps of 30 degree and note down V_0 & I_0 in tabular column
- 5) Use scope to observe input and output voltages and currents.
- 6) Save the file with a suitable name and set the run time in the menu bar to 0.1sec
- 7) Run the simulation and observe the waveforms in the scope.
- 8) Measure the value of output quantities using measurement options in the scope.
- 9) Note down the values in tabular column.

Tabular Column:

Sl. No.	Firing angle (α)	Load current (I_0)	Output voltage (V_0)

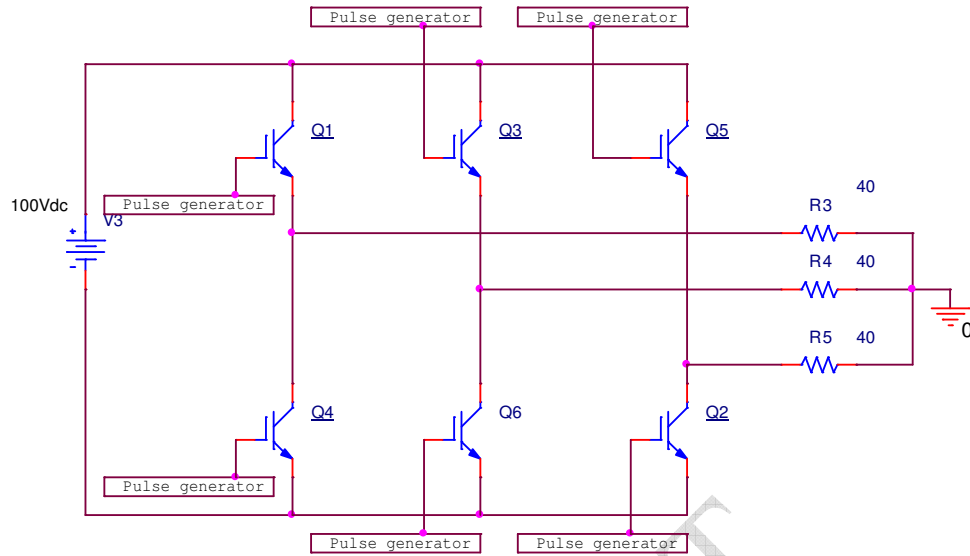
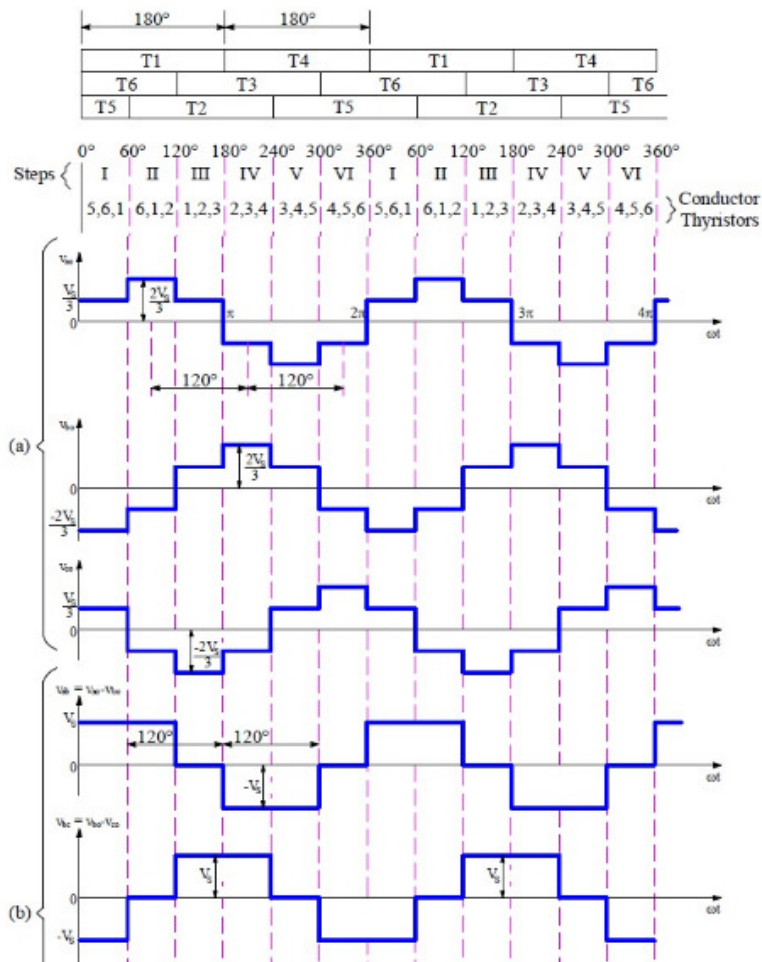
Result:

Conclusion:

Expt. No.6

Simulation of Three phase Inverter with 180° conduction

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Circuit Diagram:Nature of waveform:

Expt.No.6

Simulation of three phase inverter with 180° conduction

AIM:

Simulate a three phase inverter with 180 degree conduction mode with DC input of 100V and a star connected balanced resistive of 40 ohms each. Use IGBT for inverter.

THEORY:

PROCEDURE:

- 1) Open Matlab tool using icon on the desktop and click on Simulink icon in the home tab.
- 2) Click on new blank module and select the required components from the library function and drop on the work space.
- 3) Connect all components as per the circuit diagram
- 4) Set pulse generator with 50% pulse width and set the phase delay from 0° to 300° with increasing delay of 60° for switch Q1 to Q6.
- 5) Use scope to observe input and output voltages and currents.
- 6) Save the file with a suitable name and set the run time in the menu bar to 0.1sec
- 7) Run the simulation and observe the waveforms in the scope.
- 8) Measure the value of output quantities using measurement options in the scope.
- 9) Note down the values in tabular column.

Tabular Column:

Phase Voltage (V_{ph})	Line voltage (V_L)

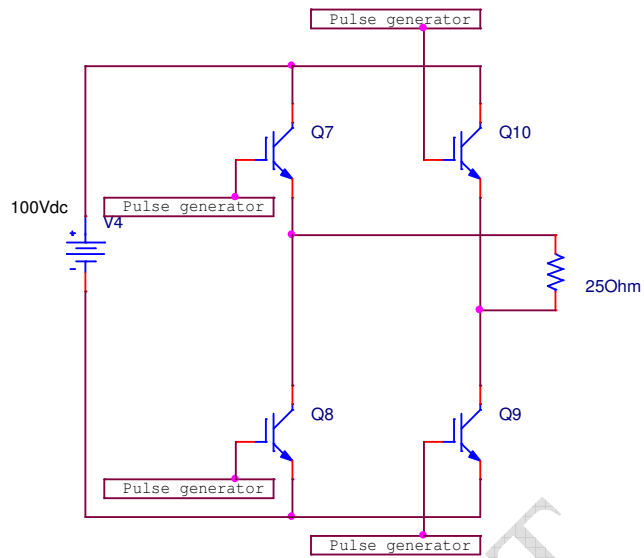
Result:

Conclusion:

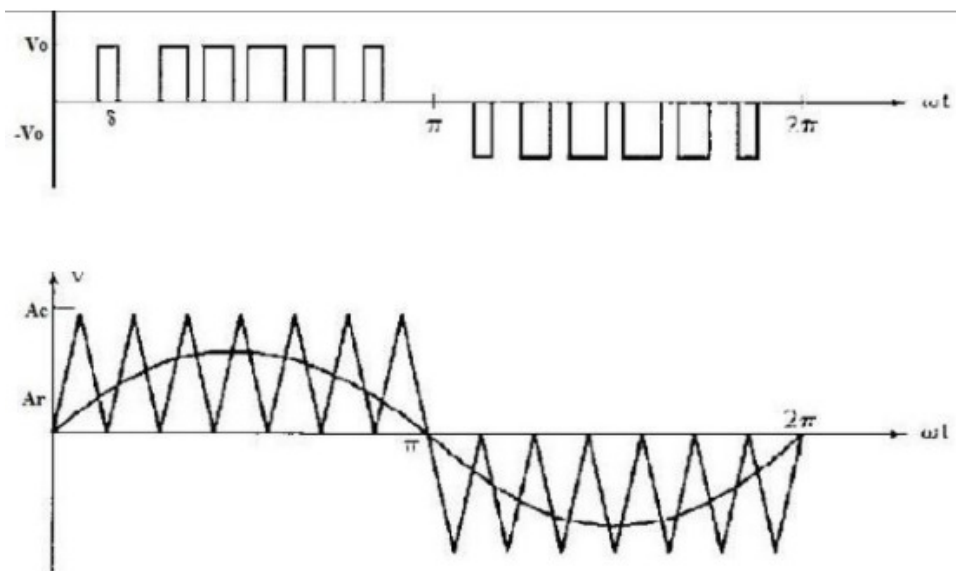
Expt. No.7

Simulation of single phase SPWM inverter

Circuit Diagram:



Nature of waveform:



Expt.No.6

Simulation of single phase SPWM inverter

AIM:

Simulate a single phase SPWM inverter with 50V DC input with modulation indices of 0.5, 0.6 and 0.8. Connect a resistance of 25 ohms at the output of inverter. Use power Mosfets for inverter.

THEORY:

PROCEDURE:

- 1) Open Matlab tool using icon on the desktop and click on Simulink icon in the home tab.
- 2) Click on new blank module and select the required components from the library function and drop on the work space.
- 3) Connect all components as per the circuit diagram
- 4) Generate SPWM pulses using comparator and relational operator.
- 5) Vary the modulation index from 0.5 to 1 in steps of 0.1 and note down the output voltage
- 6) Save the file with a suitable name and set the run time in the menu bar to 0.1sec
- 7) Run the simulation and observe the waveforms in the scope.
- 8) Measure the value of output quantities using measurement options in the scope.
- 9) Note down the values in tabular column.

Tabular Column:

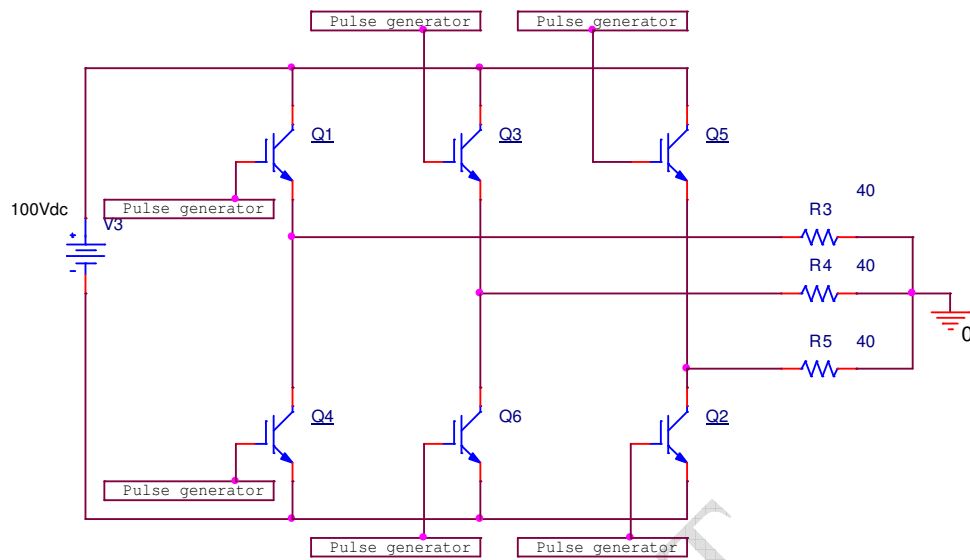
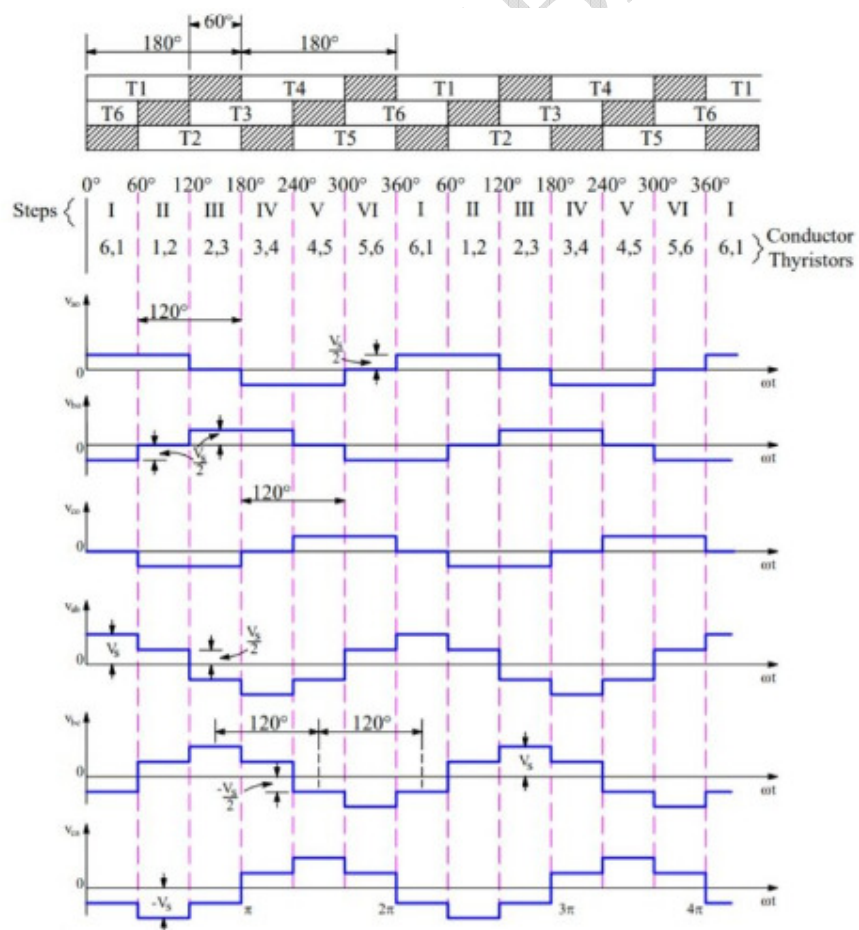
Sl. No.	Modulation Index (M)	Phase Voltage (V_{ph})	Line voltage (V_L)

Result:

Conclusion:

Expt. No.8

Simulation of Three phase Inverter with 120° conduction

Circuit Diagram:Nature of waveform:

Expt.No.8

Simulation of three phase inverter with 120° conduction

AIM:

Simulate a three phase inverter with 120 degree mode of conduction. Take input DC voltage of 100V and three phase star connected balanced resistive load of 50 ohms each.

THEORY:

PROCEDURE:

- 1) Open Matlab tool using icon on the desktop and click on Simulink icon in the home tab.
- 2) Click on new blank module and select the required components from the library function and drop on the work space.
- 3) Connect all components as per the circuit diagram
- 4) Set pulse generator with $(100/3)\%$ pulse width and set the phase delay from 0° to 300° with increasing delay of 60° for switch Q1 to Q6.
- 5) Use scope to observe input and output voltages and currents.
- 6) Save the file with a suitable name and set the run time in the menu bar to 0.1sec
- 7) Run the simulation and observe the waveforms in the scope.
- 8) Measure the value of output quantities using measurement options in the scope.
- 9) Note down the values in tabular column.

Tabular Column:

Phase Voltage (V_{ph})	Line voltage (V_L)

Result:

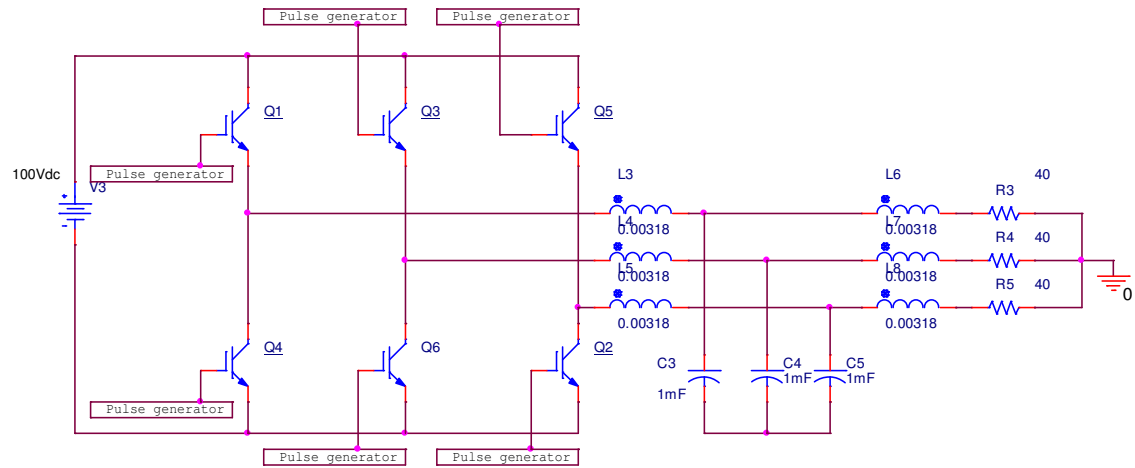
Conclusion:

Demonstration Experiments

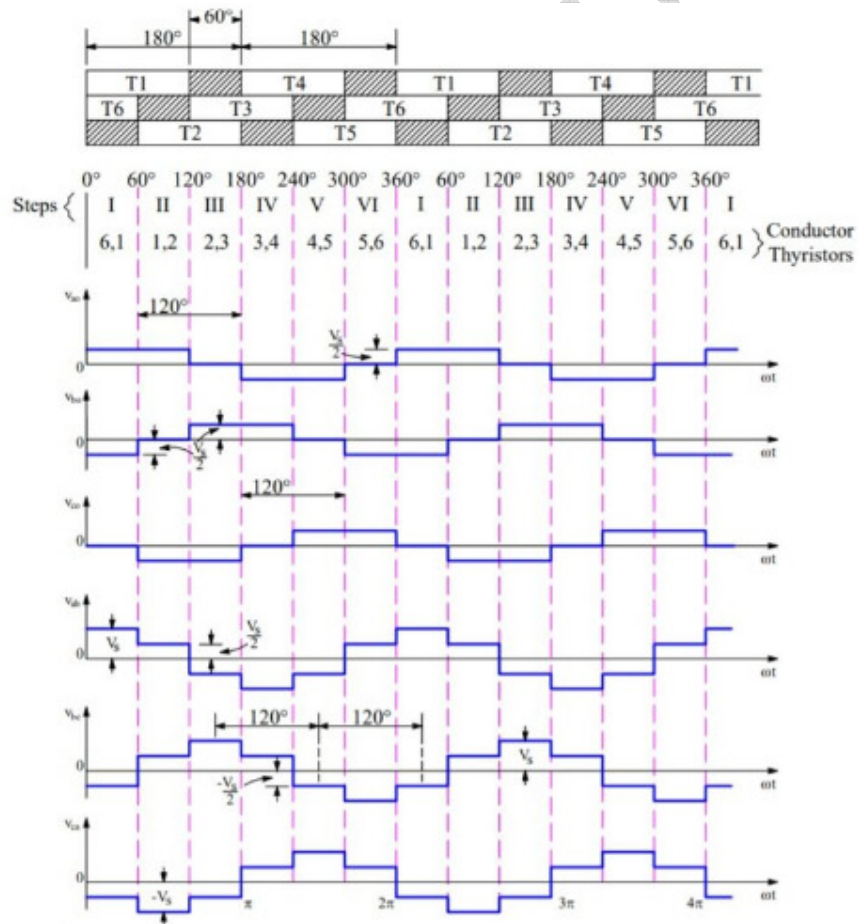
Expt. No.9

Simulation of Three phase Inverter with 120° conduction with LC filter

Circuit Diagram:



Nature of waveform:



Expt.No.9

Simulation of three phase inverter with 120° conduction with LC filter

AIM:

In expt. 8. Connect suitable LC filter at the output to obtain a sinusoidal output with THD of less than 8 %.

THEORY:

PROCEDURE:

- 1) Open Matlab tool using icon on the desktop and click on Simulink icon in the home tab.
- 2) Click on new blank module and select the required components from the library function and drop on the work space.
- 3) Connect all components as per the circuit diagram
- 4) Set pulse generator with (100/3)% pulse width and set the phase delay from 0° to 300° with increasing delay of 60° for switch Q1 to Q6.
- 5) Use scope to observe input and output voltages and currents.
- 6) Save the file with a suitable name and set the run time in the menu bar to 0.1sec
- 7) Run the simulation and observe the waveforms in the scope.
- 8) Measure the value of output quantities using measurement options in the scope.
- 9) Note down the values in tabular column.

Tabular Column:

Phase Voltage (V _{ph})	Line voltage (V _L)

Result:

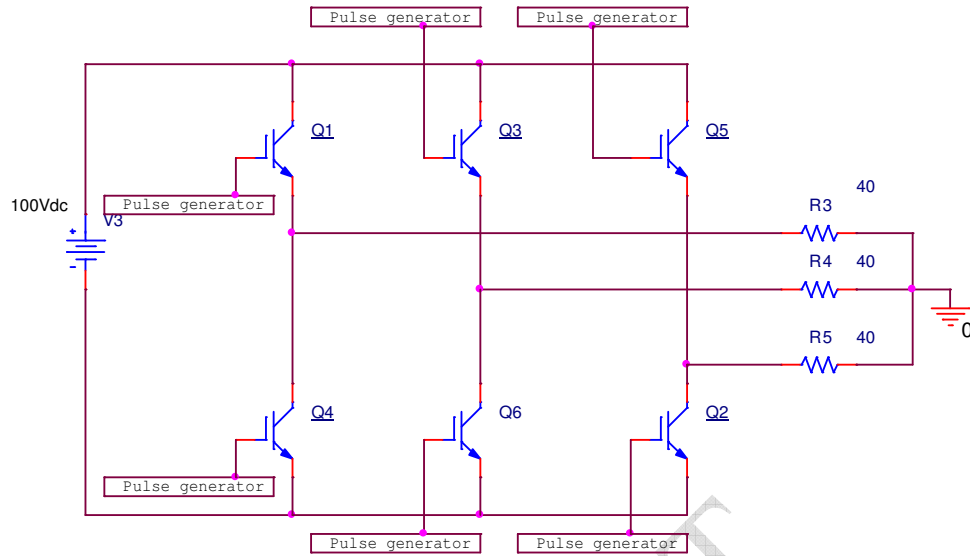
Conclusion:

Demonstration Experiments

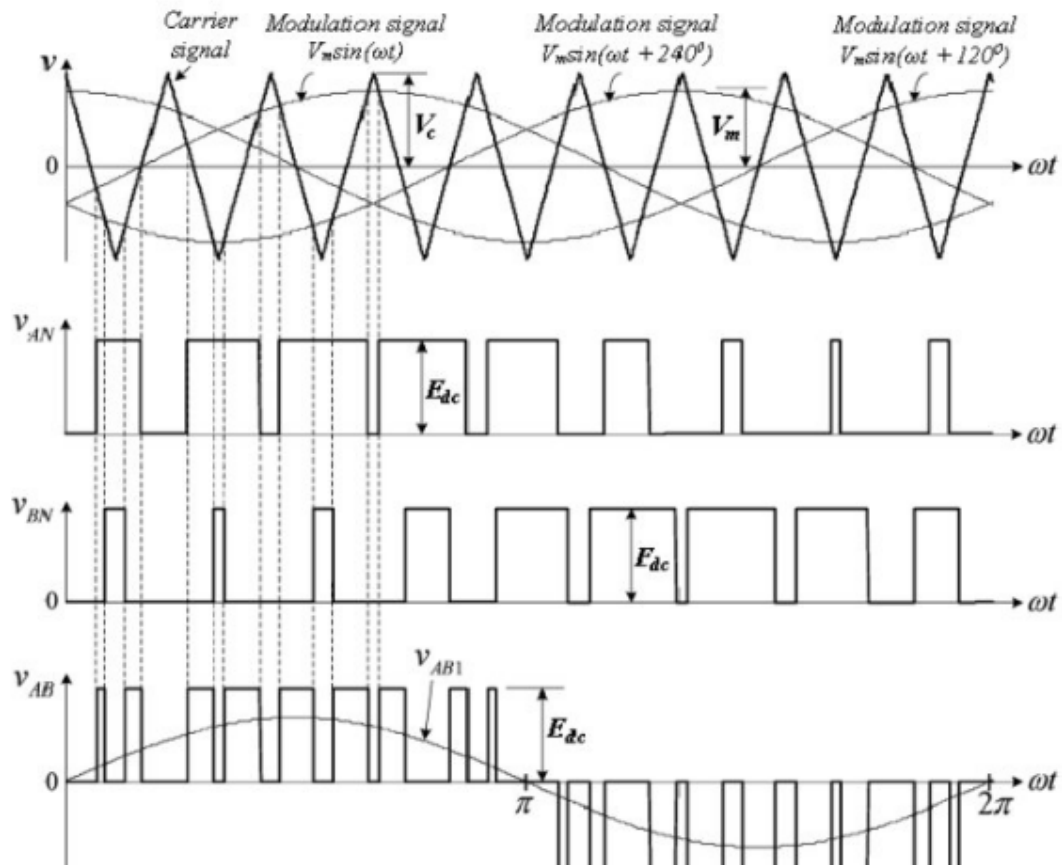
Expt. No.10

Simulation of Three phase Inverter with SPWM

Circuit Diagram:



Nature of waveform:



Expt.No.10

Simulation of three phase inverter with SPWM

AIM:

Simulate a three phase SPWM inverter with 50V DC input with modulation indices of 0.5, 0.6 and 0.8. Connect a star connected resistances of 25 ohms each at the output of the inverter. Use power Mosfets for inverter.

THEORY:

PROCEDURE:

- 1) Open Matlab tool using icon on the desktop and click on Simulink icon in the home tab.
- 2) Click on new blank module and select the required components from the library function and drop on the work space.
- 3) Connect all components as per the circuit diagram
- 4) Generate SPWM pulses using comparator adder & logical operator.
- 5) Vary the modulation index from 0.5 to 1 in steps of 0.1 and note down the output voltage.
- 6) Use scope to observe input and output voltages and currents.
- 7) Save the file with a suitable name and set the run time in the menu bar to 0.1sec
- 8) Run the simulation and observe the waveforms in the scope.
- 9) Measure the value of output quantities using measurement options in the scope.
- 10) Note down the values in tabular column.

Tabular Column:

Sl. No.	Modulation Index (M)	Phase Voltage (V_{ph})	Line voltage (V_L)

Result:

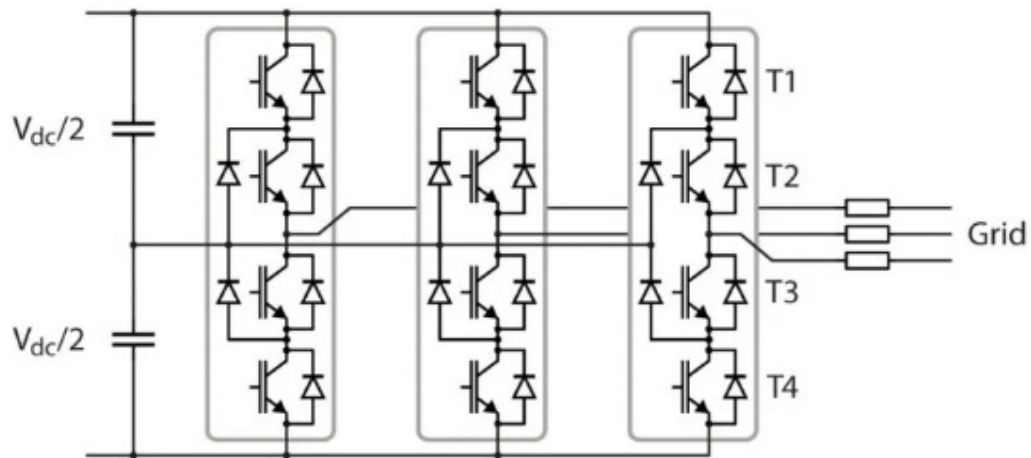
Conclusion:

Demonstration Experiments

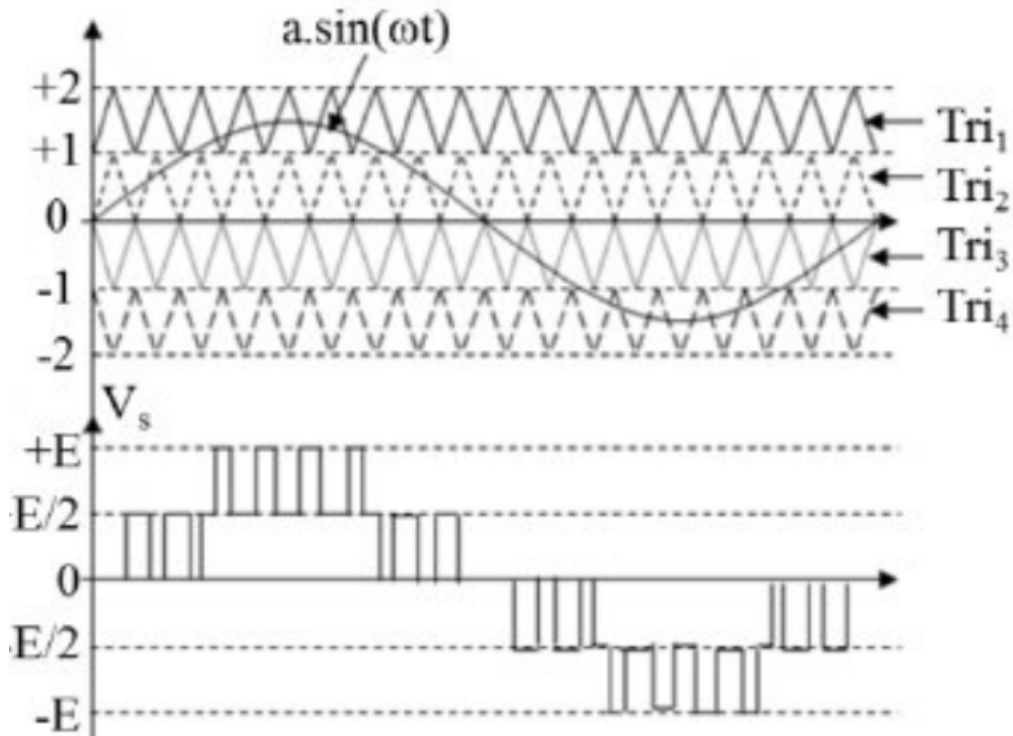
Expt. No.11

Simulation of Three phase 5 level Neutral Point Clamped (NPC) inverter

Circuit Diagram:



Nature of Waveform:



Expt.No.11

Simulation of Three phase 5 level Neutral Point Clamped (NPC) inverter

AIM:

Simulate a three phase, 5 level, neutral point clamped (NPC) inverter. Input DC voltage is 100V. The inverter output is connected to a balanced 3 phase resistive load of 40 Ohms each.

THEORY:

PROCEDURE:

- 1) Open Matlab tool using icon on the desktop and click on Simulink icon in the home tab.
- 2) Click on new blank module and select the required components from the library function and drop on the work space.
- 3) Connect all components as per the circuit diagram
- 4) Use scope to observe input and output voltages and currents.
- 5) Save the file with a suitable name and set the run time in the menu bar to 0.1sec
- 6) Run the simulation and observe the waveforms in the scope.
- 7) Measure the value of output quantities using measurement options in the scope.
- 8) Note down the values in tabular column.

Tabular Column:

Phase Voltage (V_{ph})	Line voltage (V_L)

Result:

Conclusion:

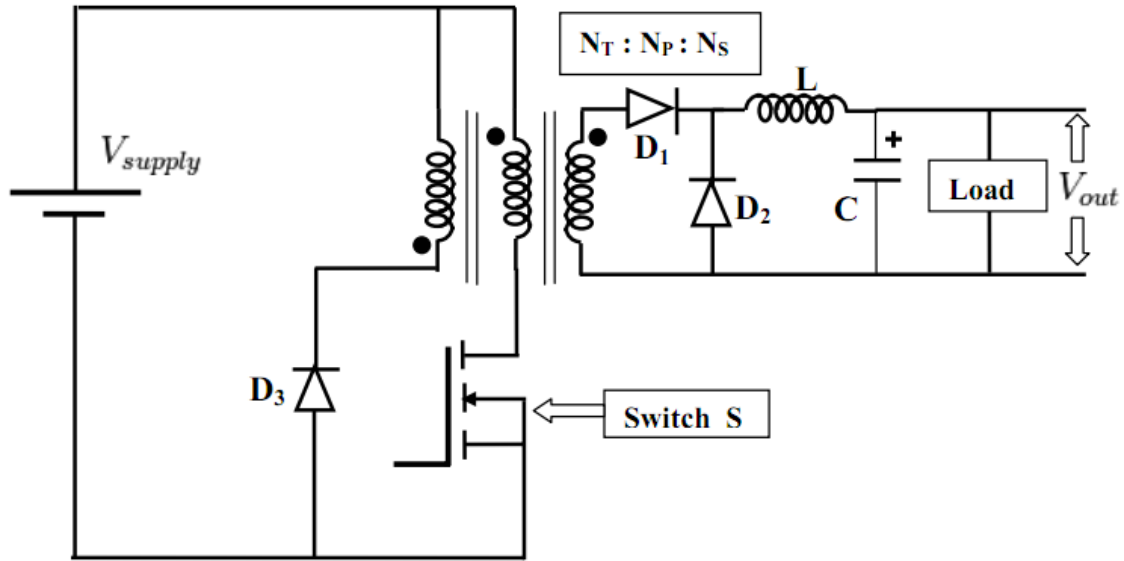
Demonstration Experiments

Expt. No.12

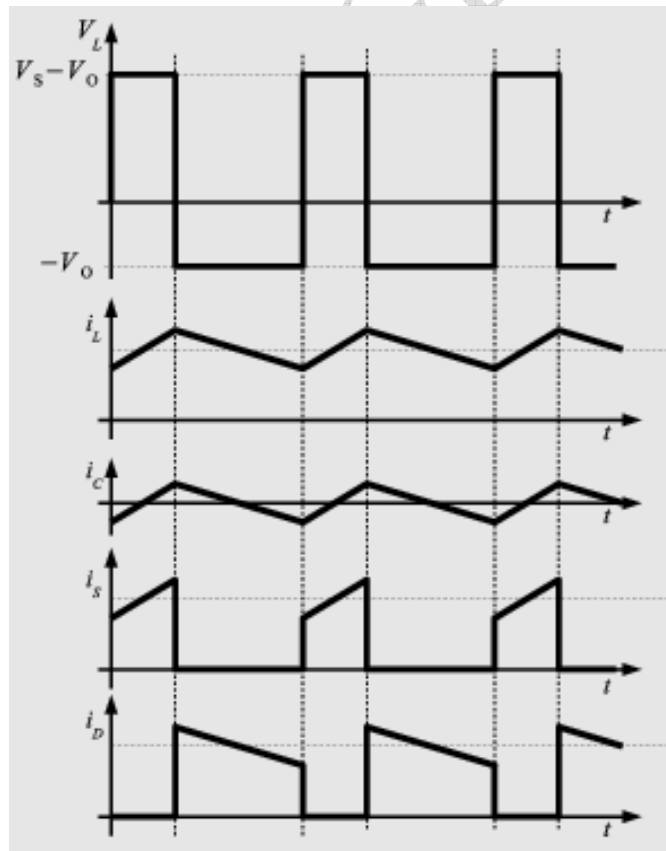
Simulation of forward converter

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Circuit Diagram:



Nature of Waveform:



Expt.No.12

Simulation of forward converter

AIM:

Simulate a forward converter with input DC voltage of 30 V. Take transformer ratio of 1.5:1. Observe the output voltages for duty cycles of 0.4, 0.6 and 0.8. Ensure that the output voltage ripple is less than 0.5 V. The load resistance is 10 Ohms.

THEORY:

PROCEDURE:

- 1) Open Matlab tool using icon on the desktop and click on Simulink icon in the home tab.
- 2) Click on new blank module and select the required components from the library function and drop on the work space.
- 3) Connect all components as per the circuit diagram
- 4) Use scope to observe input and output voltages and currents.
- 5) Save the file with a suitable name and set the run time in the menu bar to 0.1sec
- 6) Run the simulation and observe the waveforms in the scope.
- 7) Measure the value of output quantities using measurement options in the scope.
- 8) Note down the values in tabular column.

Tabular Column:

Sl. No.	Duty cycle (D)	Output Voltage (V _o)	Output Current (I _L)

Result:

Conclusion:

Simulation of control of power electronic circuits (BEEL657B)

Viva questions

1. **Definition of Power Electronics?**
2. **What is holding current in SCR?**
3. **What is latching current in SCR?**
4. **What are the different turn on methods of SCR?**
5. **What is snubber circuit?**
6. **What is hard switching of the thyristor?**
7. **What is firing angle?**
8. **What is meant by commutation?**
9. **What are the advantages of freewheeling diode in rectifier circuit?**
10. **What is meant by cyclo-converter?**
11. **What are the types of cyclo-converters?**
12. **What is ac voltage controller?**
13. **What is inverter?**
14. **What are the types of inverter?**
15. **What is duty cycle?**
16. **Can fuses with an ac voltage rating be used in a dc applications?**
17. **What are the different operation regions of the SCR?**
18. **What are the different losses that occur in thyristor while operating?**
19. **What are the advantages of speed control using thyristor?**
20. **Explain the working principal of the circuit breaker?**
21. **What is the difference between isolator and circuit breaker?**
22. **What are the advantage of freewheeling diode in a full wave rectifier?**
23. **What is chopper?**
24. **What are the types of commutation? (Or) turn off methods?**
25. **What is step down chopper?**
26. **Name some of the current controlled (current driven) devices?**
27. **Name some of the voltage driven (voltage controlled) devices.**
28. **What are the advantages of IGBT over BJT, MOSFET?**
29. **Why IGBT is very popular nowadays?**
30. **What is the difference between power diode and signal diode?**
31. **What are the different types of power MOSFET?**
32. **What is meant by phase controlled rectifier?**
33. **Mention some of the applications of controlled rectifier?**
34. **What is meant by delay angle?**
35. **What is meant by input power factor in controlled rectifier?**
36. **What are the applications of dc chopper?**
37. **What is meant by duty-cycle?**
38. **What are the main classification of inverter?**
39. **Why Thyristors are not preferred for inverters?**
40. **What are the advantages of PWM control?**

- 41. What are the methods of reduction of harmonic content?**
- 42. Compare CSI and VSI?**
- 43. What are the applications of ac voltage controllers?**
- 44. What are the two methods of control in ac voltage controllers?**
- 45. What are the two types of cyclo-converters?**
- 46. Draw the two transistor model of scr and derive an expression for anode current?**
- 47. What does the Voltmeter in AC mode show? Is it RMS value or peak value?**
- 48. What is Universal Motor?**
- 49. What are the main components used for isolating the Power Circuits, Power Semiconductor from the low-power circuit?**
- 50. Give some examples of power electronics applications in the day-to-day life?**

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